

If you are making an assumption, please write it down next to your answer so we can offer full or partial credit.

Section I: (30 points) Single Choices: select one best answer

1) Which of the following is the binary representation of integer 4141_{10} in 16bit.

- a) 0b 0001 0000 0010 0011
- b) 0b 0001 0000 0010 1101
- c) 0b 1001 0010 0100 1001
- d) 0b 0001 0000 0111 0011

2) Which of the following is the single FP representation of 4141.125.

- a) 0x 45815010
- b) 0x 45BA0A00
- c) 0x 45816B00 (* Although the real value is 45816900, this is the closest one)
- d) 0x 45815100

3) for (i=0; i<10000; i++)
 for (j=0; j<10000; j++)
 { C[j][i] = A[j][i]+B[j][i]; }

When executing the above loop on a machine with a small L1D cache (e.g. 4KB), what type of data locality may be exhibited at the L1D cache?

- a) Spatial locality
- b) Structural locality
- c) Temporal locality
- d) None of the above

4) Which one of the following statements is always true?

- a) A larger cache always hits better
- b) A direct-mapped cache always has a lower tag overhead than the same-sized set-associative cache
- c) A direct-mapped cache always has a faster hit time than the equal-sized set associative cache
- d) A large cache block always yields a better hit rate for the same-sized caches

5) Which of the following instructions is NOT presented in RV64I?

- a) LWU
- b) LDU
- c) LHU
- d) LBU

6) If you want to design a processor for low-cost mobile devices, which of the following extensions should be added to your RV32I core?

- a) M
- b) F
- c) C
- d) D

7) Which one of the following statements about VA (Virtual Address) and PA (Physical Address) is NOT true?

- a) The addresses generated by the Load/Store instructions are VA.
- b) The addresses used to access the caches are PA
- c) The page table that maps VA to PA is stored in the disk

- d) TLB is a cache for the page table
- 8) My computer has a two-way set associative cache with a 4KB and 16B line size. Which of the following memory reference sequences would run into conflict misses?
- a) 0x01ABC000, 0x01ABC004, 0x01ABC010, 0x01ABC202,
 - b) 0x01ABC000, 0x01ABC120, 0x01ABCA30, 0x01ABCD40
 - c) 0x01001000, 0x01001004, 0x00100A0, 0x01AAA000
 - d) 0x01ABA0A0, 0x01ABB0A0, 0x01ABC0AA, 0x01ABD0AA
- 9) Procedure calls are usually expensive because of the registers saving/restoring overhead. In which of the following cases does the procedure call require no such saving/restoring cost?
- a) A leaf routine that uses only callee save registers
 - b) A leaf routine that uses only caller save registers
 - c) A leaf routine that uses no registers, all variables are allocated on a stack
 - d) A non-leaf routine
- 10) For a single cycle data path design, if the D-cache miss rate is 5%, 50% of instructions are loaded/stored, and the memory latency is 100 cycles. What would be the average CPI?
- a) 2.5
 - b) 3.5
 - c) 4.5
 - d) 3

Section II (40 points): Multiple Choices (zero, one, or more correct answers)

- 1) What are the general RISC hardware design principles?
- a) Simplicity favors regularity
 - b) Make common cases fast
 - c) Keep it Simple and Stupid (KISS)
 - d) Smaller is faster
 - e) Good design demands good compromises
- 2) What are typically defined in a calling convention?
- a) Argument passing through registers
 - b) Caller/callee save registers
 - c) Function variable naming
 - d) Return value placement
 - e) Legal data types of arguments
- 3) If a 32-bit memory word consists entirely of zeros, what could this signify?
- a) Integer value 0
 - b) Floating point value +0.0
 - c) It is a denormalized number
 - d) It is four characters "null-null-null-null"
- 4) Which of the following caches are usually fully associative?
- a) A TLB
 - b) A branch predictor
 - c) L1D
 - d) A victim cache.

- 5) Virtual address aliasing occurs when two virtual addresses from different processes or even the same processes map to the same physical address. This phenomenon complicates the implementation of what structures?
 - a) A multi-level indexing page table
 - b) A virtual addressed (virtual address tagged) cache
 - c) An inverted page table.
 - d) A TLB
- 6) Which of the following mapping tables do not have tags?
 - a) A single level page table
 - b) A branch predictor
 - c) A L3 data cache
 - d) A TLB
- 7) When we increase the cache size, which of the following cache misses would decrease?
 - a) Compulsory misses
 - b) Conflict misses
 - c) Capacity misses
 - d) Coherence misses
- 8) Which of the following are often useful in reducing cache capacity misses?
 - a) Increase the cache sizes
 - b) Increase the cache block (line) sizes
 - c) Add one victim cache
 - d) Change to set associative cache
 - e) Cache prefetching
- 9) In your program, procedure A calls procedure B within a loop. The execution profile indicates that many instruction cache misses occurred for both procedures A and B. How can this situation be improved?
 - a) To turn off the instruction cache
 - b) Increase the associativity of the data cache
 - c) Inline procedure B into A, using a compiler pragma (directive)
 - d) Ask the linker to layout procedures A and B next to each other.
- 10) Which of the following memory mappings are managed by the compiler?
 - a) Memory to registers
 - b) Memory to cache
 - c) Disk to memory
 - d) Memory to local memory (LM) or the shared memory in NVidia's SM (Steaming Multiprocessor)

Section III (48 points): Short and Brief Answers

1) ISA Design Issues (10 points)

- a) Is it good to support 1024 registers in the CPU architecture? You must elaborate on your answer, not just a simple yes or no.
- b) Is it good to support the post-increment load instruction? For example, LW.post t1, 4(t2). This instruction does two actions: load data from location 4(t2) into register t1 and then increment register t2 by offset 4.
- c) Is it good to set Conditional Code (CC) as the side effect of an ALU operation rather than using the compare and branch instructions directly.

No, longer instructions, opcode and imm is squeezed; against 'Smaller is faster' principle

No, more complex data/control more pipeline hazard, not simplistic

No, need more structures, more data dependency so more data hazards, harder context switching and pipelining.

2) Data Formats (6 points)

What are the standard formats used to represent the following types of data?

- a) Signed integers **2's complement**
- b) Single and double precision floating point numbers **IEEE 754-1985**
- c) English characters **ASCII**
- d) International characters **Unicode**
- e) Half-precision floating point numbers (FP16) **IEEE754-2008, as 1, 5, 10**
- f) FP8 (mini-float) and HP8 (half-float) **as 1, 4, 3**

3) Pipelining (8 points)

- a) What hazards must be managed in a pipelined data path implementation? **Control, Data, Structural**
- b) List strategies to minimize the penalties incurred by each hazard.

1. Branch Prediction (BTB, BHT...); 2. Prefetching, Code Scheduling, Forwarding; 3. More resources (I,D-cache separation)

4) Cache organization (8 points)

Suppose a cache is organized as a 2-way set associative, with 64 sets and a block size of 16 bytes. The memory address for the current data item is 0x1234.

- a) What is the cache size? **2KB**
- b) Calculate the set selection index for the current data item. **0x1234 >> log2(16), then take the lower log2(64) bits, which is 0x23, or 35**
- c) What if the set already holds cache lines from 0x1230 and 0x4230? What will happen? **Cache hit. a line is 16b, or 0x10, so 0x1234 is in the line 0x1230.**
- d) What if the set already holds cache lines from 0x7230 and 0x4230? What will happen? **Conflict Miss. Need to evict a line to read 0x1234 into cache.**

5) Parallelism (16 points)

The following Table lists four types of parallelism often exploited in modern microprocessors

ILP	Instruction
TLP	Thread
DLP	Data
MLP	Memory

- a) What are those acronyms **(4 points)**?
- b) The following table lists many architecture/micro-architecture features exploited in modern microprocessors. For each of them, indicate which type of parallelism in the above Table this feature would support or is related to **(12 points)**:

Branch predictors ILP	Snoopy Caches TLP
Out-of-order execution MLP, ILP	Vector instructions DLP
Non-blocking caches MLP	GPU DLP, MLP, TLP
Pipelining ILP	Sub-word parallel instructions DLP
Multi-cores TLP	Inclusive Caches TLP
MSHR MLP	Interleaved memory banks MLP